Sun Yat-sen University (SYSU)
Course Title: Pattern Recognition
Instructors:

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Department: School of Computer Science and Engineering
Credits: 3 credits (54 contact hours over 18 weeks; 3 contact hours per week)
Semester Offered: Spring 2023
Course Page: https://csai-sysu.net/aicourse/HTML/home.html

Course Description

This graduate-level course is designed for master students in information disciplines at SYSU (with emphasis on Computer Science and Technology). It gives a systematic introduction to the theoretical foundations and practical techniques of **Pattern Recognition**, while keeping track of fast-developing AI technologies.

The course is structured in three layers:

1. **Foundations of statistical pattern recognition**: Bayesian decision theory, linear and nonlinear classifiers, neural networks, HMMs, deep sequence models, optimization, ensemble learning, unsupervised and weakly supervised learning.
2. **Representation and compression for intelligent perception**: dimensionality reduction, model compression, and task-oriented feature learning.
3. **Applications in language, vision, video, and data mining**: natural language processing basics, image understanding, video analysis and understanding, and an introduction to recommender systems.

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Explain the mathematical framework of Bayesian decision theory and apply it to pattern classification.
2. Distinguish between generative and discriminative modeling, and implement linear and nonlinear classifiers.
3. Describe core components of neural networks and deep learning, including network optimization and sequence models.
4. Model temporal and structured data with HMMs and modern deep sequence models.
5. Apply ensemble, unsupervised, and weak/weakly-supervised learning methods to real-world data.
6. Use dimensionality reduction and model compression ideas to make models more deployable.
7. Understand basic tasks and pipelines in NLP, image understanding, video analysis, and recommender systems.
8. Conduct independent literature review and write a structured technical survey.
9. Work in a small research group to complete an English, CVPR-style project paper on one of the given research-oriented topics, and present it.

Course Topics

Module	Topic	Key Concepts / Methods
1	Overview of Artificial Intelligence	AI history, application panorama, perception vs cognition, course organization
2	Bayesian Foundations and Classification	Bayesian decision theory, risk, loss, MAP and ML estimation
3	Linear and Nonlinear Classifiers	Discriminant functions, geometric interpretation, capacity and regularization
4	Artificial Neural Networks	Perceptron, MLP, activation, loss, optimization, intro to CNN
5	HMM and Markov Models	Markov chains, HMM three problems, EM/Baum–Welch
6	Deep Sequence Models	RNN, LSTM/GRU, attention, encoder–decoder
7	Ensemble Learning and Network Optimization	Bagging, boosting, random forest, error–diversity view
8	Unsupervised Learning	Clustering, similarity, representation, applications

9	Generalized Weakly Supervised Learning	Incomplete/inexact/inaccurate supervision, semi-supervised learning, co-training
10	Dimensionality Reduction and Model Compression	PCA, manifold learning ideas, random projection, motivation for compression
11	Basics of NLP	NLP pipeline, word representations, sequence labeling, dialogue/Q&A basics
12	Computer Vision: Image Feature Extraction and Understanding	Hand-crafted features (SIFT/HOG/LBP), BoVW, deep visual features, detection/segmentation
13	Computer Vision: Video Analysis – Feature Extraction	Spatio-temporal interest points, dense trajectory, two-stream / 3D CNN ideas
14	Computer Vision: Video Understanding & Vision–Language	Action recognition/detection, tracking, video object segmentation, video captioning
15	Data Mining for Recommender Systems	Content-based and collaborative filtering, latent factor models, explainable recommendation
16–19	Project Presentation & Course Wrap-up	Group presentations, Q&A, discussion on challenges and trends

(Actual week-to-topic mapping may be slightly adjusted according to progress.)

Course Assessment

Overall Grading Scheme

Component	**Description**	**Weight**
Individual Assignment	Survey report (≥ 5000 words), independent work	**40%**
Group Project	3–4 students, research-oriented project, CVPR template, English	**50%**
Participation / In-class Performance	Attendance, sign-in, class discussion, cold calls	**10%**

1. Individual Assignment (40%)

- **Task**: Choose one specific topic from the course (e.g., weakly supervised learning, multi-modal video understanding, domain adaptation, robust visual recognition, deep sequence models) and write an **introductory survey**.
- **Content requirements**:

    - Problem / topic definition
    - Taxonomy of models / methods
    - Core ideas and pros/cons of representative techniques
    - Recent advances and SOTA trends
    - Open challenges and research directions
    - Proper figures / illustrations
    - **Standard references** with correct citation format
- **Length**: main text **not less than 5000 characters/words**
- **Format**: Word document, filename "Name-Student ID", includes reference list and states actual word count
- **Deadline**: **End of Week 18**

2. Group Project (50%)

- **Group size**: 3–4 students
- **Goal**: Pick **one** topic from the 6 official research-oriented project topics (see below), conduct literature review, propose or adapt a method, run experiments on public datasets, and write it up like a short conference paper.
- **Report format**:
    - **English**
    - **CVPR double-column template**
    - Sections: Title, Abstract, Introduction, Method, Experiments, Conclusion, References
    - **Main text ≥ 4 pages** (references not counted)
    - Must state **contribution ratio** of each member
    - **Deadline**: end of **Week 19**
- **Presentation**: Weeks 15–18: every group presents; **each member must answer at least one question.**
- **Mid-term check (Week 12)**: submit progress report (Word) including:
    - Group members and task division
    - Introduction + Related Work for chosen topic
    - At least one baseline method run on at least one dataset
    - Current difficulties and key problems

    - Planned solutions
    - References follow GB/T style as in the course notice.

Official Group Project Topics (choose 1):

1. **Multimodal Video–Language Analysis** (PIC challenge tasks)
2. **Few-shot Image Classification** (medical + 2 natural image datasets; compare with ≥ 5 methods)
3. **Noisy-label Robust Visual Recognition** (image classification / face recognition; improve over existing methods, e.g., CVPR'21 augmentation strategy)
4. **Universal Domain Adaptation for Visual Recognition** (VISDA-2021; improve over https://arxiv.org/abs/2104.03344)
5. **Community Detection in Complex Networks with High-order Structures** (design new high-order features; ≥ 5% accuracy improvement)
6. **Dialogue Management for Multi-turn Human–Machine Conversation** (DSTC8 Track 2, Ubuntu or Advising; propose better model and report leaderboard-style results)

Grading for the group project will consider: (i) **paper writing quality**, (ii) **methodological novelty**, (iii) **experimental completeness / comparison**, and (iv) **presentation & Q&A performance**.

3. Participation / In-class Performance (10%)

- Unannounced roll calls
- In-class answers / discussions
- Overall engagement with the course

Course Policies

- **Academic integrity**: Individual assignment must be completed independently; plagiarism or excessive copying from survey/tutorial papers will result in zero for that component.
- **Deadlines**: Late submission without an approved reason may be penalized.
- **Teamwork**: For the group project, contribution ratio must be clearly stated; instructors reserve the right to adjust individual scores within a group.